

ABSTRACT

This report presents the design and implementation of a desktop-based Hostel Management System developed using Java Swing, JDBC, and MySQL. The primary objective of the system is to simplify the management of hostel accommodations by providing an efficient and user-friendly interface for maintaining student and room allocation records. The system reduces manual effort, improves record organization, and enhances the overall management process within hostel facilities.

The application supports essential operations such as adding, viewing, updating, and deleting student hostel records through a graphical user interface. Each record contains information such as student name, room number, block name, and occupancy status. The system is structured using a layered architecture consisting of Data Transfer Object (DTO), Data Access Object (DAO), Service, and User Interface (UI) layers. This separation of concerns promotes modularity, simplifies maintenance, and enables future scalability of the application.

The DAO layer is responsible for handling database operations using JDBC, ensuring secure and efficient communication with the MySQL database. The service layer incorporates business logic and input validation to maintain the accuracy and consistency of hostel data. The UI layer, developed using Java Swing, provides an intuitive platform for users to interact with the system and perform operations seamlessly.

This project demonstrates the integration of a Java-based graphical interface with a relational database system and highlights the importance of structured software design in developing reliable, maintainable, and user-friendly hostel management applications.